Expanded Grade 4 Math Practice

Volumes 1 and 2 | 8 Chapters | 240 Original Questions | Visual Lessons and Worked Solutions

This workbook follows the Grade 4 unit scope you supplied. Every lesson, visual, question, answer, and explanation is newly written practice material and is not copied from a textbook.

Contents

Volume 1

Chapters 1-4: whole-number operations and algebraic thinking.

Chapter 1: Place Value and Multi-Digit Addition and Subtraction

Concept: Reading, writing, and rounding large numbers

What it means: Every digit has a place value. Moving one place left makes its value 10 times greater. **Example:** In 582,641, the 8 is worth 80,000. To round 582,641 to the nearest ten thousand, look at the thousands digit, 2. Since 2 is less than 5, the answer is 580,000. **Remember:** Find the target place, look one digit to the right, and then round.

Visual model

Millions	Hundred-thousands	Ten-thousands	Thousands
5	8	2	6

Questions

1. Round 582,641 to the nearest ten thousand.

2. Round 769,328 to the nearest hundred thousand.

3. Round 431,876 to the nearest thousand.

4. Round 915,492 to the nearest ten thousand.

5. Round 246,751 to the nearest hundred thousand.

6. Round 680,149 to the nearest thousand.

7. Round 357,862 to the nearest ten thousand.

8. Round 824,399 to the nearest hundred thousand.

9. Round 508,741 to the nearest thousand.

10. Round 693,258 to the nearest ten thousand.

Concept: Adding multi-digit numbers

What it means: Put digits with the same place value in vertical columns. Regroup when a column adds to 10 or more. **Example:** In 4,786 + 2,549, add ones first. When the ones total 15, write 5 ones and regroup 1 ten. **Remember:** Begin on the right and move left one place at a time.

Visual model

$$\begin{array}{r} 4{,}786 \\ +\ 2{,}549 \\ \hline 7{,}335 \end{array}$$

Questions

11. Find 34,786 + 25,449.

12. Find 38,003 + 27,483.

13. Find 41,220 + 29,517.

14. Find 44,437 + 31,551.

15. Find 47,654 + 33,585.

16. Find 50,871 + 35,619.

17. Find 54,088 + 37,653.

18. Find 57,305 + 39,687.

19. Find 60,522 + 41,721.

20. Find 63,739 + 43,755.

Concept: Subtracting multi-digit numbers

What it means: Subtraction compares or takes away amounts. Regroup when the top digit is smaller than the bottom digit. **Example:** In 6,203 - 1,478, trade 1 hundred for 10 tens, then continue as needed. **Remember:** Check by adding your difference to the number you subtracted.

Visual model

$$\begin{array}{r} 4{,}786 \\ -\ 1{,}478 \\ \hline 4{,}725 \end{array}$$

Questions

21. Find 86,203 - 41,478.

22. Find 88,722 - 42,785.

23. Find 91,241 - 44,092.

24. Find 93,760 - 45,399.

25. Find 96,279 - 46,706.

26. Find 98,798 - 48,013.

27. Find 101,317 - 49,320.

28. Find 103,836 - 50,627.

29. Find 106,355 - 51,934.

30. Find 108,874 - 53,241.

Chapter Answers and Explained Solutions

Use these explanations to understand the concept and check how the answer is found.

1.1 Answer: **580,000**
Look at the digit to the right of the ten thousand place. Round 582,641 to 580,000.

1.2 Answer: **800,000**
Look at the digit to the right of the hundred thousand place. Round 769,328 to 800,000.

1.3 Answer: **432,000**
Look at the digit to the right of the thousand place. Round 431,876 to 432,000.

1.4 Answer: **920,000**
Look at the digit to the right of the ten thousand place. Round 915,492 to 920,000.

1.5 Answer: **200,000**
Look at the digit to the right of the hundred thousand place. Round 246,751 to 200,000.

1.6 Answer: **680,000**
Look at the digit to the right of the thousand place. Round 680,149 to 680,000.

1.7 Answer: **360,000**
Look at the digit to the right of the ten thousand place. Round 357,862 to 360,000.

1.8 Answer: **800,000**
Look at the digit to the right of the hundred thousand place. Round 824,399 to 800,000.

1.9 Answer: **509,000**
Look at the digit to the right of the thousand place. Round 508,741 to 509,000.

1.10 Answer: **690,000**
Look at the digit to the right of the ten thousand place. Round 693,258 to 690,000.

1.11 Answer: **60,235**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 34,786 + 25,449 = 60,235.

1.12 Answer: **65,486**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 38,003 + 27,483 = 65,486.

1.13 Answer: **70,737**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 41,220 + 29,517 = 70,737.

1.14 Answer: **75,988**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 44,437 + 31,551 = 75,988.

1.15 Answer: **81,239**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 47,654 + 33,585 = 81,239.

1.16 Answer: **86,490**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 50,871 + 35,619 = 86,490.

1.17 Answer: **91,741**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 54,088 + 37,653 = 91,741.

1.18 Answer: **96,992**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 57,305 + 39,687 = 96,992.

1.19 Answer: **102,243**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 60,522 + 41,721 = 102,243.

1.20 Answer: **107,494**
Line up place values and add from right to left, regrouping when a column totals 10 or more: 63,739 + 43,755 = 107,494.

1.21 Answer: **44,725**
Line up place values and subtract from right to left. Regroup when needed: 86,203 - 41,478 = 44,725.

1.22 Answer: **45,937**
Line up place values and subtract from right to left. Regroup when needed: 88,722 - 42,785 = 45,937.

1.23 Answer: **47,149**
Line up place values and subtract from right to left. Regroup when needed: 91,241 - 44,092 = 47,149.

1.24 Answer: **48,361**
Line up place values and subtract from right to left. Regroup when needed: 93,760 - 45,399 = 48,361.

1.25 Answer: **49,573**
Line up place values and subtract from right to left. Regroup when needed: 96,279 - 46,706 = 49,573.

1.26 Answer: **50,785**
Line up place values and subtract from right to left. Regroup when needed: 98,798 - 48,013 = 50,785.

1.27 Answer: **51,997**
Line up place values and subtract from right to left. Regroup when needed: 101,317 - 49,320 = 51,997.

1.28 Answer: **53,209**
Line up place values and subtract from right to left. Regroup when needed: 103,836 - 50,627 = 53,209.

1.29 Answer: **54,421**
Line up place values and subtract from right to left. Regroup when needed: 106,355 - 51,934 = 54,421.

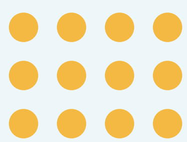
1.30 Answer: **55,633**
Line up place values and subtract from right to left. Regroup when needed: 108,874 - 53,241 = 55,633.

Chapter 2: Multiplication with Whole Numbers

Concept: Arrays and area models

What it means: An array has equal rows and columns. An area model splits a larger factor into tens, hundreds, and ones. **Example:** For 23 x 4, split 23 into 20 + 3. Then 20 x 4 = 80 and 3 x 4 = 12, so the product is 92. **Remember:** Add every partial product.

Visual model



23 x 4

20 x 4 + 3 x 4 = 92

Questions

1. Use an area model to find 23 x 3.

2. Use an area model to find 30 x 4.

3. Use an area model to find 37 x 5.

4. Use an area model to find 44 x 6.

5. Use an area model to find 51 x 7.

6. Use an area model to find 58 x 8.

7. Use an area model to find 65 x 9.

8. Use an area model to find 72 x 10.

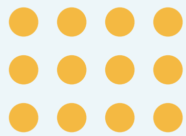
9. Use an area model to find 79 x 11.

10. Use an area model to find 86 x 12.

Concept: Multiplying a 1-digit number by a 4-digit number

What it means: Multiply each place value by the one-digit factor, regrouping when needed. **Example:** For 3 x 1,426, multiply 3 by ones, tens, hundreds, and thousands in order. **Remember:** Estimate first. Since 3 x 1,400 is about 4,200, a product near 4,278 is reasonable.

Visual model



23 x 4

20 x 4 + 3 x 4 = 92

Questions

11. Find 1,426 x 2.

12. Find 1,743 x 3.

13. Find 2,060 x 4.

14. Find 2,377 x 5.

15. Find 2,694 x 6.

16. Find 3,011 x 7.

17. Find 3,328 x 2.

18. Find 3,645 x 3.

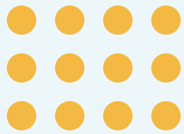
19. Find 3,962 x 4.

20. Find 4,279 x 5.

Concept: Multiplying two 2-digit numbers

What it means: Break one factor into tens and ones. **Example:** For 24 x 13, calculate 24 x 10 = 240 and 24 x 3 = 72; then add 240 + 72 = 312. **Remember:** The tens partial product represents tens, not ones.

Visual model



23 x 4

20 x 4 + 3 x 4 = 92

Questions

21. Find 12 x 14.

22. Find 13 x 16.

23. Find 14 x 18.

24. Find 15 x 20.

25. Find 16 x 22.

26. Find 17 x 24.

27. Find 18 x 26.

28. Find 19 x 28.

29. Find 20 x 30.

30. Find 21 x 32.

Chapter Answers and Explained Solutions

Use these explanations to understand the concept and check how the answer is found.

2.1 Answer: **69**
Split 23 into tens and ones, multiply each part by 3, and add the partial products to get 69.

2.2 Answer: **120**
Split 30 into tens and ones, multiply each part by 4, and add the partial products to get 120.

2.3 Answer: **185**
Split 37 into tens and ones, multiply each part by 5, and add the partial products to get 185.

2.4 Answer: **264**
Split 44 into tens and ones, multiply each part by 6, and add the partial products to get 264.

2.5 Answer: **357**
Split 51 into tens and ones, multiply each part by 7, and add the partial products to get 357.

2.6 Answer: **464**
Split 58 into tens and ones, multiply each part by 8, and add the partial products to get 464.

2.7 Answer: **585**
Split 65 into tens and ones, multiply each part by 9, and add the partial products to get 585.

2.8 Answer: **720**
Split 72 into tens and ones, multiply each part by 10, and add the partial products to get 720.

2.9 Answer: **869**
Split 79 into tens and ones, multiply each part by 11, and add the partial products to get 869.

2.10 Answer: **1032**
Split 86 into tens and ones, multiply each part by 12, and add the partial products to get 1032.

2.11 Answer: **2,852**
Multiply 1,426 by 2 one place at a time, regrouping when needed. The product is 2,852.

2.12 Answer: **5,229**
Multiply 1,743 by 3 one place at a time, regrouping when needed. The product is 5,229.

2.13 Answer: **8,240**
Multiply 2,060 by 4 one place at a time, regrouping when needed. The product is 8,240.

2.14 Answer: **11,885**
Multiply 2,377 by 5 one place at a time, regrouping when needed. The product is 11,885.

2.15 Answer: **16,164**
Multiply 2,694 by 6 one place at a time, regrouping when needed. The product is 16,164.

2.16 Answer: **21,077**
Multiply 3,011 by 7 one place at a time, regrouping when needed. The product is 21,077.

2.17 Answer: **6,656**
Multiply 3,328 by 2 one place at a time, regrouping when needed. The product is 6,656.

2.18 Answer: **10,935**
Multiply 3,645 by 3 one place at a time, regrouping when needed. The product is 10,935.

2.19 Answer: **15,848**
Multiply 3,962 by 4 one place at a time, regrouping when needed. The product is 15,848.

2.20 Answer: **21,395**
Multiply 4,279 by 5 one place at a time, regrouping when needed. The product is 21,395.

2.21 Answer: **168**
Break 14 into tens and ones: 12 x 10 + 12 x 4 = 168.

2.22 Answer: **208**
Break 16 into tens and ones: 13 x 10 + 13 x 6 = 208.

2.23 Answer: **252**
Break 18 into tens and ones: 14 x 10 + 14 x 8 = 252.

2.24 Answer: **300**
Break 20 into tens and ones: 15 x 20 + 15 x 0 = 300.

2.25 Answer: **352**
Break 22 into tens and ones: 16 x 20 + 16 x 2 = 352.

2.26 Answer: **408**
Break 24 into tens and ones: 17 x 20 + 17 x 4 = 408.

2.27 Answer: **468**
Break 26 into tens and ones: 18 x 20 + 18 x 6 = 468.

2.28 Answer: **532**
Break 28 into tens and ones: 19 x 20 + 19 x 8 = 532.

2.29 Answer: **600**
Break 30 into tens and ones: 20 x 30 + 20 x 0 = 600.

2.30 Answer: **672**
Break 32 into tens and ones: 21 x 30 + 21 x 2 = 672.

Chapter 3: Division with Whole Numbers

Concept: Division vocabulary and equal groups

What it means: The dividend is the amount being shared, the divisor tells how many equal groups, and the quotient is the amount in each group. **Example:** In 84 divided by 6 = 14, 84 is the dividend, 6 is the divisor, and 14 is the quotient. **Remember:** Multiply divisor x quotient to check.

Visual model

4) 864 **= 216**

divisor x quotient = dividend

Questions

1. Name the dividend, divisor, and quotient in 84 divided by 6 = 14.

2. Name the dividend, divisor, and quotient in 96 divided by 8 = 12.

3. Name the dividend, divisor, and quotient in 72 divided by 9 = 8.

4. Name the dividend, divisor, and quotient in 63 divided by 7 = 9.

5. Name the dividend, divisor, and quotient in 144 divided by 12 = 12.

6. Name the dividend, divisor, and quotient in 120 divided by 10 = 12.

7. Name the dividend, divisor, and quotient in 132 divided by 11 = 12.

8. Name the dividend, divisor, and quotient in 108 divided by 9 = 12.

9. Name the dividend, divisor, and quotient in 156 divided by 12 = 13.

10. Name the dividend, divisor, and quotient in 175 divided by 7 = 25.

Concept: Long division with a 1-digit divisor

What it means: Long division repeats four steps: divide, multiply, subtract, and bring down. **Example:** To divide 864 by 4, work from the hundreds place to the ones place. **Remember:** A quotient digit belongs directly above the place you divided.

Visual model

4) 864 **= 216**

divisor x quotient = dividend

Questions

11. Divide 864 by 3. Give a quotient and remainder.

12. Divide 989 by 4. Give a quotient and remainder.

13. Divide 1,114 by 5. Give a quotient and remainder.

14. Divide 1,239 by 6. Give a quotient and remainder.

15. Divide 1,364 by 7. Give a quotient and remainder.

16. Divide 1,489 by 8. Give a quotient and remainder.

17. Divide 1,614 by 3. Give a quotient and remainder.

18. Divide 1,739 by 4. Give a quotient and remainder.

19. Divide 1,864 by 5. Give a quotient and remainder.

20. Divide 1,989 by 6. Give a quotient and remainder.

Concept: Interpreting remainders

What it means: A remainder is what is left after making equal groups. The story decides what to do with it. **Example:** If 29 students ride vans that hold 6 students, 29 divided by 6 is 4 remainder 5. You need 5 vans, because the 5 students still need a ride. **Remember:** A remainder is always smaller than the divisor.

Visual model

4) 864 = 216

divisor x quotient = dividend

Questions

21. 29 students need vans with 6 seats each. How many vans are needed?

22. 47 students need vans with 8 seats each. How many vans are needed?

23. 73 students need vans with 10 seats each. How many vans are needed?

24. 58 students need vans with 9 seats each. How many vans are needed?

25. 65 students need vans with 12 seats each. How many vans are needed?

26. 94 students need vans with 15 seats each. How many vans are needed?

27. 38 students need vans with 7 seats each. How many vans are needed?

28. 53 students need vans with 8 seats each. How many vans are needed?

29. 76 students need vans with 11 seats each. How many vans are needed?

30. 101 students need vans with 20 seats each. How many vans are needed?

Chapter Answers and Explained Solutions

Use these explanations to understand the concept and check how the answer is found.

3.1 Answer: **Dividend 84; divisor 6; quotient 14**
The dividend is being shared (84), the divisor tells the group size (6), and the quotient is the answer (14).

3.2 Answer: **Dividend 96; divisor 8; quotient 12**
The dividend is being shared (96), the divisor tells the group size (8), and the quotient is the answer (12).

3.3 Answer: **Dividend 72; divisor 9; quotient 8**
The dividend is being shared (72), the divisor tells the group size (9), and the quotient is the answer (8).

3.4 Answer: **Dividend 63; divisor 7; quotient 9**
The dividend is being shared (63), the divisor tells the group size (7), and the quotient is the answer (9).

3.5 Answer: **Dividend 144; divisor 12; quotient 12**
The dividend is being shared (144), the divisor tells the group size (12), and the quotient is the answer (12).

3.6 Answer: **Dividend 120; divisor 10; quotient 12**
The dividend is being shared (120), the divisor tells the group size (10), and the quotient is the answer (12).

3.7 Answer: **Dividend 132; divisor 11; quotient 12**
The dividend is being shared (132), the divisor tells the group size (11), and the quotient is the answer (12).

3.8 Answer: **Dividend 108; divisor 9; quotient 12**
The dividend is being shared (108), the divisor tells the group size (9), and the quotient is the answer (12).

3.9 Answer: **Dividend 156; divisor 12; quotient 13**
The dividend is being shared (156), the divisor tells the group size (12), and the quotient is the answer (13).

3.10 Answer: **Dividend 175; divisor 7; quotient 25**
The dividend is being shared (175), the divisor tells the group size (7), and the quotient is the answer (25).

3.11 Answer: **288 R 0**
Use divide, multiply, subtract, and bring down. Check: 3 x 288 + 0 = 864.

3.12 Answer: **247 R 1**
Use divide, multiply, subtract, and bring down. Check: 4 x 247 + 1 = 989.

3.13 Answer: **222 R 4**
Use divide, multiply, subtract, and bring down. Check: 5 x 222 + 4 = 1,114.

3.14 Answer: **206 R 3**
Use divide, multiply, subtract, and bring down. Check: 6 x 206 + 3 = 1,239.

3.15 Answer: **194 R 6**
Use divide, multiply, subtract, and bring down. Check: 7 x 194 + 6 = 1,364.

3.16 Answer: **186 R 1**
Use divide, multiply, subtract, and bring down. Check: 8 x 186 + 1 = 1,489.

3.17 Answer: **538 R 0**
Use divide, multiply, subtract, and bring down. Check: 3 x 538 + 0 = 1,614.

3.18 Answer: **434 R 3**
Use divide, multiply, subtract, and bring down. Check: 4 x 434 + 3 = 1,739.

3.19 Answer: **372 R 4**
Use divide, multiply, subtract, and bring down. Check: 5 x 372 + 4 = 1,864.

3.20 Answer: **331 R 3**
Use divide, multiply, subtract, and bring down. Check: 6 x 331 + 3 = 1,989.

3.21 Answer: **5**
29 divided by 6 has a remainder, so one more van is needed for the remaining students. The answer is 5 vans.

3.22 Answer: **6**
47 divided by 8 has a remainder, so one more van is needed for the remaining students. The answer is 6 vans.

3.23 Answer: **8**
73 divided by 10 has a remainder, so one more van is needed for the remaining students. The answer is 8 vans.

3.24 Answer: **7**
58 divided by 9 has a remainder, so one more van is needed for the remaining students. The answer is 7 vans.

3.25 Answer: **6**
65 divided by 12 has a remainder, so one more van is needed for the remaining students. The answer is 6 vans.

3.26 Answer: **7**
94 divided by 15 has a remainder, so one more van is needed for the remaining students. The answer is 7 vans.

3.27 Answer: **6**
38 divided by 7 has a remainder, so one more van is needed for the remaining students. The answer is 6 vans.

3.28 Answer: **7**
53 divided by 8 has a remainder, so one more van is needed for the remaining students. The answer is 7 vans.

3.29 Answer: **7**
76 divided by 11 has a remainder, so one more van is needed for the remaining students. The answer is 7 vans.

3.30 Answer: **6**
101 divided by 20 has a remainder, so one more van is needed for the remaining students. The answer is 6 vans.

Chapter 4: Equations and Word Problems

Concept: Solving multi-step word problems

What it means: Multi-step problems need more than one operation. Read carefully, organize the facts, solve one step, and use that result in the next step. **Example:** If 6 boxes hold 24 pencils each and 17 pencils are given away, first multiply, then subtract. **Remember:** Write an equation for each step and label the answer.

Visual model

$$n + 36 = 91 \qquad n = 91 - 36 = 55$$

Use an inverse operation to find n

Questions

1. A shop has 3 boxes with 12 pencils each. It gives away 5 pencils. How many pencils remain?

2. A shop has 4 boxes with 17 pencils each. It gives away 9 pencils. How many pencils remain?

3. A shop has 5 boxes with 22 pencils each. It gives away 13 pencils. How many pencils remain?

4. A shop has 6 boxes with 27 pencils each. It gives away 17 pencils. How many pencils remain?

5. A shop has 7 boxes with 32 pencils each. It gives away 21 pencils. How many pencils remain?

6. A shop has 8 boxes with 37 pencils each. It gives away 25 pencils. How many pencils remain?

7. A shop has 9 boxes with 42 pencils each. It gives away 29 pencils. How many pencils remain?

8. A shop has 10 boxes with 47 pencils each. It gives away 33 pencils. How many pencils remain?

9. A shop has 11 boxes with 52 pencils each. It gives away 37 pencils. How many pencils remain?

10. A shop has 12 boxes with 57 pencils each. It gives away 41 pencils. How many pencils remain?

Concept: Variables and equations

What it means: A variable is a letter that stands for an unknown number. An equation says two amounts are equal. **Example:** In n + 36 = 91, subtract 36 from both sides to find n = 55. **Remember:** Use the inverse operation to undo what happened to the variable.

Visual model

$$n + 36 = 91 \qquad n = 91 - 36 = 55$$

Use an inverse operation to find n

Questions

11. Solve n + 25 = 90.

12. Solve n + 28 = 97.

13. Solve n + 31 = 104.

14. Solve n + 34 = 111.

15. Solve n + 37 = 118.

16. Solve n + 40 = 125.

17. Solve n + 43 = 132.

18. Solve n + 46 = 139.

19. Solve n + 49 = 146.

20. Solve n + 52 = 153.

Concept: Factors, multiples, prime, and composite numbers

What it means: Factors multiply to make a number. Multiples are the numbers made by multiplying. A prime number has exactly two factors; a composite number has more than two. **Example:** 18 has factors 1, 2, 3, 6, 9, and 18, so it is composite. **Remember:** 1 is neither prime nor composite.

Visual model

$$n + 36 = 91 \qquad n = 91 - 36 = 55$$

Use an inverse operation to find n

Questions

21. List the factors of 20. Then decide whether it is prime or composite.

22. List the factors of 21. Then decide whether it is prime or composite.

23. List the factors of 22. Then decide whether it is prime or composite.

24. List the factors of 23. Then decide whether it is prime or composite.

25. List the factors of 24. Then decide whether it is prime or composite.

26. List the factors of 25. Then decide whether it is prime or composite.

27. List the factors of 26. Then decide whether it is prime or composite.

28. List the factors of 27. Then decide whether it is prime or composite.

29. List the factors of 28. Then decide whether it is prime or composite.

30. List the factors of 29. Then decide whether it is prime or composite.

Chapter Answers and Explained Solutions

Use these explanations to understand the concept and check how the answer is found.

4.1 Answer: **31**
First find the total: 3 x 12 = 36. Then subtract 5: 36 - 5 = 31.

4.2 Answer: **59**
First find the total: 4 x 17 = 68. Then subtract 9: 68 - 9 = 59.

4.3 Answer: **97**
First find the total: 5 x 22 = 110. Then subtract 13: 110 - 13 = 97.

4.4 Answer: **145**
First find the total: 6 x 27 = 162. Then subtract 17: 162 - 17 = 145.

4.5 Answer: **203**
First find the total: 7 x 32 = 224. Then subtract 21: 224 - 21 = 203.

4.6 Answer: **271**
First find the total: 8 x 37 = 296. Then subtract 25: 296 - 25 = 271.

4.7 Answer: **349**
First find the total: 9 x 42 = 378. Then subtract 29: 378 - 29 = 349.

4.8 Answer: **437**
First find the total: 10 x 47 = 470. Then subtract 33: 470 - 33 = 437.

4.9 Answer: **535**
First find the total: 11 x 52 = 572. Then subtract 37: 572 - 37 = 535.

4.10 Answer: **643**
First find the total: 12 x 57 = 684. Then subtract 41: 684 - 41 = 643.

4.11 Answer: **65**
Subtract 25 from both sides: n = 90 - 25 = 65.

4.12 Answer: **69**
Subtract 28 from both sides: n = 97 - 28 = 69.

4.13 Answer: **73**
Subtract 31 from both sides: n = 104 - 31 = 73.

4.14 Answer: **77**
Subtract 34 from both sides: n = 111 - 34 = 77.

4.15 Answer: **81**
Subtract 37 from both sides: n = 118 - 37 = 81.

4.16 Answer: **85**
Subtract 40 from both sides: n = 125 - 40 = 85.

4.17 Answer: **89**
Subtract 43 from both sides: n = 132 - 43 = 89.

4.18 Answer: **93**
Subtract 46 from both sides: n = 139 - 46 = 93.

4.19 Answer: **97**
Subtract 49 from both sides: n = 146 - 49 = 97.

4.20 Answer: **101**
Subtract 52 from both sides: n = 153 - 52 = 101.

4.21 Answer: **Factors: 1, 2, 4, 5, 10, 20; composite**
Factors divide evenly into 20. It has 6 factors, so it is composite.

4.22 Answer: **Factors: 1, 3, 7, 21; composite**
Factors divide evenly into 21. It has 4 factors, so it is composite.

4.23 Answer: **Factors: 1, 2, 11, 22; composite**
Factors divide evenly into 22. It has 4 factors, so it is composite.

4.24 Answer: **Factors: 1, 23; prime**
Factors divide evenly into 23. It has 2 factors, so it is prime.

4.25 Answer: **Factors: 1, 2, 3, 4, 6, 8, 12, 24; composite**
Factors divide evenly into 24. It has 8 factors, so it is composite.

4.26 Answer: **Factors: 1, 5, 25; composite**
Factors divide evenly into 25. It has 3 factors, so it is composite.

4.27 Answer: **Factors: 1, 2, 13, 26; composite**
Factors divide evenly into 26. It has 4 factors, so it is composite.

4.28 Answer: **Factors: 1, 3, 9, 27; composite**
Factors divide evenly into 27. It has 4 factors, so it is composite.

4.29 Answer: **Factors: 1, 2, 4, 7, 14, 28; composite**
Factors divide evenly into 28. It has 6 factors, so it is composite.

4.30 Answer: **Factors: 1, 29; prime**
Factors divide evenly into 29. It has 2 factors, so it is prime.

Volume 2

Chapters 5-8: measurement, fractions, decimals, and geometry.

Chapter 5: Measurement

Concept: Converting measurement units

What it means: A conversion changes the unit name but not the amount. **Example:** Since 1 foot equals 12 inches, 4 feet equals 4 x 12 = 48 inches. **Remember:** When changing from a larger unit to a smaller unit, the number gets larger.

Visual model

4 feet **x 12** **48 inches**

1 foot = 12 inches

Questions

1. Convert 3 hours to minutes.

2. Convert 4 feet to inches.

3. Convert 7 yards to feet.

4. Convert 5 meters to centimeters.

5. Convert 2 kilograms to grams.

6. Convert 6 pints to cups.

7. Convert 3 quarts to pints.

8. Convert 2 gallons to quarts.

9. Convert 9 days to hours.

10. Convert 5 weeks to days.

Concept: Time, metric weight, and liquid volume

What it means: Use conversion facts to solve with time, grams and kilograms, or cups, pints, quarts, and gallons. **Example:** Since 1 kilogram is 1,000 grams, 3 kilograms is 3,000 grams. **Remember:** Write the conversion fact before you calculate.

Visual model

4 feet **x 12** **48 inches**

1 foot = 12 inches

Questions

11. A bag has 2 kilograms of rice. How many grams is that?

12. A bag has 3 kilograms of rice. How many grams is that?

13. A bag has 4 kilograms of rice. How many grams is that?

14. A bag has 5 kilograms of rice. How many grams is that?

15. A bag has 6 kilograms of rice. How many grams is that?

16. A bag has 7 kilograms of rice. How many grams is that?

17. A bag has 8 kilograms of rice. How many grams is that?

18. A bag has 9 kilograms of rice. How many grams is that?

19. A bag has 10 kilograms of rice. How many grams is that?

20. A bag has 11 kilograms of rice. How many grams is that?

Concept: Perimeter and area in real life

What it means: Perimeter measures the distance around an object. Area measures the space inside a flat shape. **Example:** A 7-meter by 4-meter garden has perimeter 22 meters and area 28 square meters. **Remember:** Perimeter uses units; area uses square units.

Visual model

Perimeter = 22 m; Area = 28 square m

Questions

21. A rectangle is 5 m by 3 m. Find its perimeter and area.

22. A rectangle is 6 m by 4 m. Find its perimeter and area.

23. A rectangle is 7 m by 5 m. Find its perimeter and area.

24. A rectangle is 8 m by 6 m. Find its perimeter and area.

25. A rectangle is 9 m by 7 m. Find its perimeter and area.

26. A rectangle is 10 m by 8 m. Find its perimeter and area.

27. A rectangle is 11 m by 9 m. Find its perimeter and area.

28. A rectangle is 12 m by 10 m. Find its perimeter and area.

29. A rectangle is 13 m by 11 m. Find its perimeter and area.

30. A rectangle is 14 m by 12 m. Find its perimeter and area.

Chapter Answers and Explained Solutions

Use these explanations to understand the concept and check how the answer is found.

5.1 Answer: **180 minutes**
Use the conversion fact, then multiply: 3 hours = 180 minutes.

5.2 Answer: **48 inches**
Use the conversion fact, then multiply: 4 feet = 48 inches.

5.3 Answer: **21 feet**
Use the conversion fact, then multiply: 7 yards = 21 feet.

5.4 Answer: **500 centimeters**
Use the conversion fact, then multiply: 5 meters = 500 centimeters.

5.5 Answer: **2000 grams**
Use the conversion fact, then multiply: 2 kilograms = 2000 grams.

5.6 Answer: **12 cups**
Use the conversion fact, then multiply: 6 pints = 12 cups.

5.7 Answer: **6 pints**
Use the conversion fact, then multiply: 3 quarts = 6 pints.

5.8 Answer: **8 quarts**
Use the conversion fact, then multiply: 2 gallons = 8 quarts.

5.9 Answer: **216 hours**
Use the conversion fact, then multiply: 9 days = 216 hours.

5.10 Answer: **35 days**
Use the conversion fact, then multiply: 5 weeks = 35 days.

5.11 Answer: **2,000 grams**
One kilogram is 1,000 grams, so 2 x 1,000 = 2,000 grams.

5.12 Answer: **3,000 grams**
One kilogram is 1,000 grams, so 3 x 1,000 = 3,000 grams.

5.13 Answer: **4,000 grams**
One kilogram is 1,000 grams, so 4 x 1,000 = 4,000 grams.

5.14 Answer: **5,000 grams**
One kilogram is 1,000 grams, so 5 x 1,000 = 5,000 grams.

5.15 Answer: **6,000 grams**
One kilogram is 1,000 grams, so 6 x 1,000 = 6,000 grams.

5.16 Answer: **7,000 grams**
One kilogram is 1,000 grams, so 7 x 1,000 = 7,000 grams.

5.17 Answer: **8,000 grams**
One kilogram is 1,000 grams, so 8 x 1,000 = 8,000 grams.

5.18 Answer: **9,000 grams**
One kilogram is 1,000 grams, so 9 x 1,000 = 9,000 grams.

5.19 Answer: **10,000 grams**
One kilogram is 1,000 grams, so 10 x 1,000 = 10,000 grams.

5.20 Answer: **11,000 grams**
One kilogram is 1,000 grams, so 11 x 1,000 = 11,000 grams.

5.21 Answer: **Perimeter 16 m; area 15 square m**
Perimeter = 2 x (5 + 3) = 16 m. Area = 5 x 3 = 15 square meters.

5.22 Answer: **Perimeter 20 m; area 24 square m**
Perimeter = 2 x (6 + 4) = 20 m. Area = 6 x 4 = 24 square meters.

5.23 Answer: **Perimeter 24 m; area 35 square m**
Perimeter = 2 x (7 + 5) = 24 m. Area = 7 x 5 = 35 square meters.

5.24 Answer: **Perimeter 28 m; area 48 square m**
Perimeter = 2 x (8 + 6) = 28 m. Area = 8 x 6 = 48 square meters.

5.25 Answer: **Perimeter 32 m; area 63 square m**
Perimeter = 2 x (9 + 7) = 32 m. Area = 9 x 7 = 63 square meters.

5.26 Answer: **Perimeter 36 m; area 80 square m**
Perimeter = 2 x (10 + 8) = 36 m. Area = 10 x 8 = 80 square meters.

5.27 Answer: **Perimeter 40 m; area 99 square m**
Perimeter = 2 x (11 + 9) = 40 m. Area = 11 x 9 = 99 square meters.

5.28 Answer: **Perimeter 44 m; area 120 square m**
Perimeter = 2 x (12 + 10) = 44 m. Area = 12 x 10 = 120 square meters.

5.29 Answer: **Perimeter 48 m; area 143 square m**
Perimeter = 2 x (13 + 11) = 48 m. Area = 13 x 11 = 143 square meters.

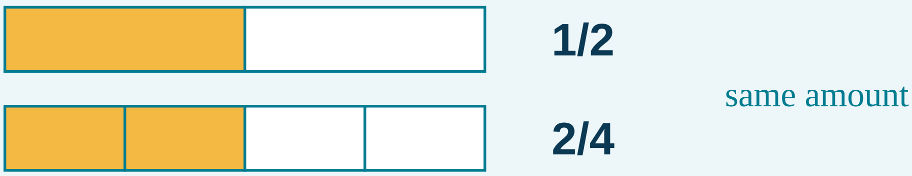
5.30 Answer: **Perimeter 52 m; area 168 square m**
Perimeter = 2 x (14 + 12) = 52 m. Area = 14 x 12 = 168 square meters.

Chapter 6: Fraction Concepts and Operations

Concept: Equivalent fractions and simplest form

What it means: Equivalent fractions name the same amount. A fraction is in simplest form when the numerator and denominator have no common factor except 1. **Example:** 2/4 = 1/2 because both numerator and denominator can be divided by 2. **Remember:** Whatever you do to the top, do to the bottom.

Visual model



Questions

1. Write 1/2 in simplest form.

2. Write 2/3 in simplest form.

3. Write 3/4 in simplest form.

4. Write 2/5 in simplest form.

5. Write 3/8 in simplest form.

6. Write 4/6 in simplest form.

7. Write 5/10 in simplest form.

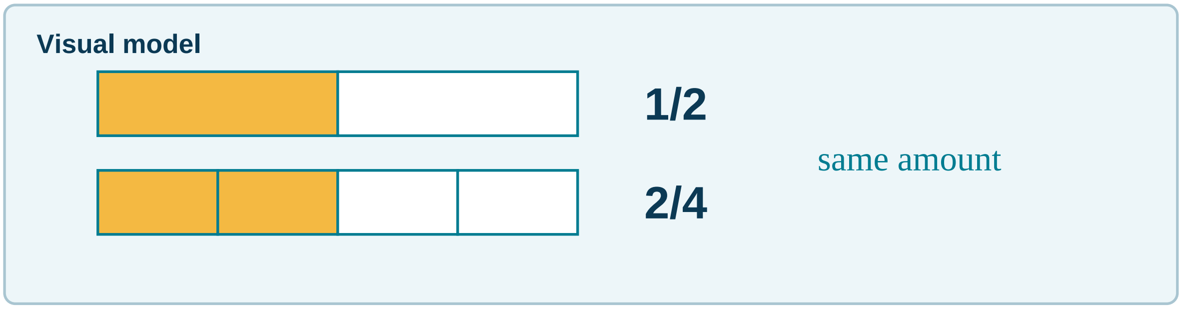
8. Write 3/9 in simplest form.

9. Write 4/12 in simplest form.

10. Write 6/8 in simplest form.

Concept: Adding fractions with like denominators

What it means: Fractions with the same denominator have equal-size pieces. Add the numerators and keep the denominator. **Example:** 3/8 + 2/8 = 5/8. **Remember:** Do not add the denominators when they already match.



Questions

11. Find 1/5 + 1/5.

12. Find 2/6 + 2/6.

13. Find 3/7 + 3/7.

14. Find 4/8 + 1/8.

15. Find 1/9 + 2/9.

16. Find 2/10 + 3/10.

17. Find 3/11 + 1/11.

18. Find 4/12 + 2/12.

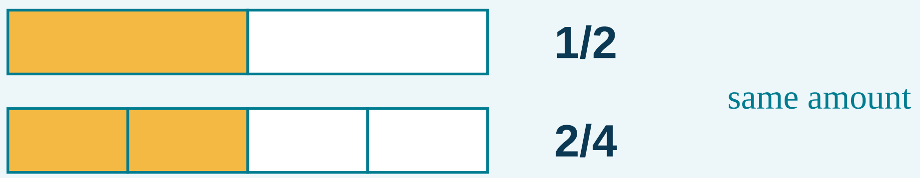
19. Find 1/13 + 3/13.

20. Find 2/14 + 1/14.

Concept: Subtracting fractions and mixed numbers

What it means: Subtract fractions with like denominators by subtracting the numerators. **Example:** 7/10 - 3/10 = 4/10 = 2/5. **Remember:** When subtracting mixed numbers, regroup one whole as fractional pieces if the fraction is too small.

Visual model



Questions

21. Find 4/7 - 1/7.

22. Find 5/8 - 2/8.

23. Find 6/9 - 3/9.

24. Find 7/10 - 1/10.

25. Find 8/11 - 2/11.

26. Find 4/12 - 3/12.

27. Find 5/13 - 1/13.

28. Find 6/14 - 2/14.

29. Find 7/15 - 3/15.

30. Find 8/16 - 1/16.

Chapter Answers and Explained Solutions

Use these explanations to understand the concept and check how the answer is found.

6.1 Answer: **1/2**
Divide the numerator and denominator by their greatest common factor to get 1/2.

6.2 Answer: **2/3**
Divide the numerator and denominator by their greatest common factor to get 2/3.

6.3 Answer: **3/4**
Divide the numerator and denominator by their greatest common factor to get 3/4.

6.4 Answer: **2/5**
Divide the numerator and denominator by their greatest common factor to get 2/5.

6.5 Answer: **3/8**
Divide the numerator and denominator by their greatest common factor to get 3/8.

6.6 Answer: **2/3**
Divide the numerator and denominator by their greatest common factor to get 2/3.

6.7 Answer: **1/2**
Divide the numerator and denominator by their greatest common factor to get 1/2.

6.8 Answer: **1/3**
Divide the numerator and denominator by their greatest common factor to get 1/3.

6.9 Answer: **1/3**
Divide the numerator and denominator by their greatest common factor to get 1/3.

6.10 Answer: **3/4**
Divide the numerator and denominator by their greatest common factor to get 3/4.

6.11 Answer: **2/5**
The denominators match, so add the numerators: 1 + 1 = 2. Keep denominator 5.

6.12 Answer: **4/6**
The denominators match, so add the numerators: 2 + 2 = 4. Keep denominator 6.

6.13 Answer: **6/7**
The denominators match, so add the numerators: 3 + 3 = 6. Keep denominator 7.

6.14 Answer: **5/8**
The denominators match, so add the numerators: 4 + 1 = 5. Keep denominator 8.

6.15 Answer: **3/9**
The denominators match, so add the numerators: 1 + 2 = 3. Keep denominator 9.

6.16 Answer: **5/10**
The denominators match, so add the numerators: 2 + 3 = 5. Keep denominator 10.

6.17 Answer: **4/11**
The denominators match, so add the numerators: 3 + 1 = 4. Keep denominator 11.

6.18 Answer: **6/12**
The denominators match, so add the numerators: 4 + 2 = 6. Keep denominator 12.

6.19 Answer: **4/13**
The denominators match, so add the numerators: 1 + 3 = 4. Keep denominator 13.

6.20 Answer: **3/14**
The denominators match, so add the numerators: 2 + 1 = 3. Keep denominator 14.

6.21 Answer: **3/7**
The denominators match, so subtract the numerators: 4 - 1 = 3. Keep denominator 7.

6.22 Answer: **3/8**
The denominators match, so subtract the numerators: 5 - 2 = 3. Keep denominator 8.

6.23 Answer: **3/9**
The denominators match, so subtract the numerators: 6 - 3 = 3. Keep denominator 9.

6.24 Answer: **6/10**
The denominators match, so subtract the numerators: 7 - 1 = 6. Keep denominator 10.

6.25 Answer: **6/11**
The denominators match, so subtract the numerators: 8 - 2 = 6. Keep denominator 11.

6.26 Answer: **1/12**
The denominators match, so subtract the numerators: 4 - 3 = 1. Keep denominator 12.

6.27 Answer: **4/13**
The denominators match, so subtract the numerators: 5 - 1 = 4. Keep denominator 13.

6.28 Answer: **4/14**
The denominators match, so subtract the numerators: 6 - 2 = 4. Keep denominator 14.

6.29 Answer: **4/15**
The denominators match, so subtract the numerators: 7 - 3 = 4. Keep denominator 15.

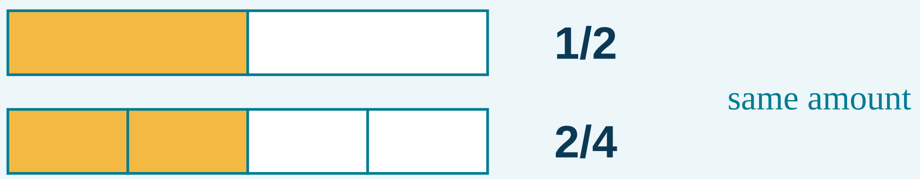
6.30 Answer: **7/16**
The denominators match, so subtract the numerators: 8 - 1 = 7. Keep denominator 16.

Chapter 7: Fractions and Decimals

Concept: Multiplying a fraction by a whole number

What it means: A whole number times a fraction means repeated groups of that fraction. **Example:** 3 x 2/5 = 6/5 = 1 1/5. **Remember:** Multiply the numerator by the whole number, then simplify or make a mixed number.

Visual model



Questions

1. Find 4 x 1/3.

2. Find 3 x 2/5.

3. Find 2 x 3/4.

4. Find 5 x 2/7.

5. Find 4 x 3/8.

6. Find 3 x 4/9.

7. Find 2 x 5/6.

8. Find 6 x 1/8.

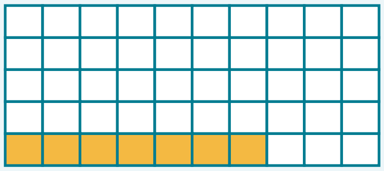
9. Find 4 x 3/5.

10. Find 3 x 7/10.

Concept: Tenths and hundredths

What it means: Decimals are another way to write fractions with denominators of 10, 100, and so on. **Example:** 0.7 is seven tenths, or 7/10. The decimal 0.35 is thirty-five hundredths, or 35/100. **Remember:** The first place right of the decimal is tenths; the second is hundredths.

Visual model



0.7 = 7/10

0.35 = 35/100

Questions

11. Write 1/10 as a decimal.

12. Write 2/10 as a decimal.

13. Write 3/10 as a decimal.

14. Write 4/10 as a decimal.

15. Write 5/10 as a decimal.

16. Write 6/10 as a decimal.

17. Write 7/10 as a decimal.

18. Write 8/10 as a decimal.

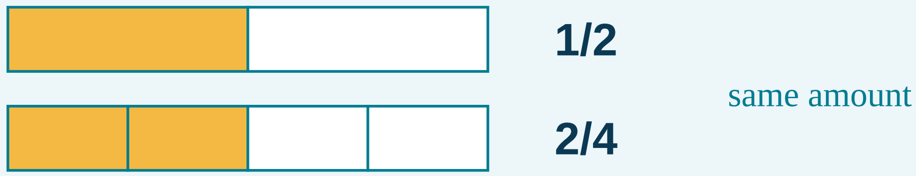
19. Write 9/10 as a decimal.

20. Write 10/10 as a decimal.

Concept: Comparing decimals and fractions

What it means: Compare quantities by writing them in a common form. **Example:** 3/4 = 0.75, so 0.75 is greater than 0.7. **Remember:** Add a zero if needed: 0.7 = 0.70.

Visual model



Questions

21. Compare 0.25 and 0.18. Use <, >, or =.

22. Compare 0.35 and 0.28. Use <, >, or =.

23. Compare 0.45 and 0.38. Use <, >, or =.

24. Compare 0.55 and 0.48. Use <, >, or =.

25. Compare 0.65 and 0.58. Use <, >, or =.

26. Compare 0.75 and 0.68. Use <, >, or =.

27. Compare 0.85 and 0.78. Use <, >, or =.

28. Compare 0.95 and 0.88. Use <, >, or =.

29. Compare 1.05 and 0.98. Use <, >, or =.

30. Compare 1.15 and 1.08. Use <, >, or =.

Chapter Answers and Explained Solutions

Use these explanations to understand the concept and check how the answer is found.

7.1 Answer: **1 1/3**
Multiply the numerator by 4: 4 x 1/3 = 4/3. Simplify or write as a mixed number: 1 1/3.

7.2 Answer: **1 1/5**
Multiply the numerator by 3: 3 x 2/5 = 6/5. Simplify or write as a mixed number: 1 1/5.

7.3 Answer: **1 2/4**
Multiply the numerator by 2: 2 x 3/4 = 6/4. Simplify or write as a mixed number: 1 2/4.

7.4 Answer: **1 3/7**
Multiply the numerator by 5: 5 x 2/7 = 10/7. Simplify or write as a mixed number: 1 3/7.

7.5 Answer: **1 4/8**
Multiply the numerator by 4: 4 x 3/8 = 12/8. Simplify or write as a mixed number: 1 4/8.

7.6 Answer: **1 3/9**
Multiply the numerator by 3: 3 x 4/9 = 12/9. Simplify or write as a mixed number: 1 3/9.

7.7 Answer: **1 4/6**
Multiply the numerator by 2: 2 x 5/6 = 10/6. Simplify or write as a mixed number: 1 4/6.

7.8 Answer: **6/8**
Multiply the numerator by 6: 6 x 1/8 = 6/8. Simplify or write as a mixed number: 6/8.

7.9 Answer: **2 2/5**
Multiply the numerator by 4: 4 x 3/5 = 12/5. Simplify or write as a mixed number: 2 2/5.

7.10 Answer: **2 1/10**
Multiply the numerator by 3: 3 x 7/10 = 21/10. Simplify or write as a mixed number: 2 1/10.

7.11 Answer: **0.1**
A denominator of 10 means tenths. Put 1 in the tenths place: 0.1.

7.12 Answer: **0.2**
A denominator of 10 means tenths. Put 2 in the tenths place: 0.2.

7.13 Answer: **0.3**
A denominator of 10 means tenths. Put 3 in the tenths place: 0.3.

7.14 Answer: **0.4**
A denominator of 10 means tenths. Put 4 in the tenths place: 0.4.

7.15 Answer: **0.5**
A denominator of 10 means tenths. Put 5 in the tenths place: 0.5.

7.16 Answer: **0.6**
A denominator of 10 means tenths. Put 6 in the tenths place: 0.6.

7.17 Answer: **0.7**
A denominator of 10 means tenths. Put 7 in the tenths place: 0.7.

7.18 Answer: **0.8**
A denominator of 10 means tenths. Put 8 in the tenths place: 0.8.

7.19 Answer: **0.9**
A denominator of 10 means tenths. Put 9 in the tenths place: 0.9.

7.20 Answer: **1.0**
A denominator of 10 means tenths. Put 10 in the tenths place: 1.0.

7.21 Answer: **>**
Compare tenths and then hundredths. 0.25 > 0.18.

7.22 Answer: **>**
Compare tenths and then hundredths. 0.35 > 0.28.

7.23 Answer: **>**
Compare tenths and then hundredths. 0.45 > 0.38.

7.24 Answer: **>**
Compare tenths and then hundredths. 0.55 > 0.48.

7.25 Answer: **>**
Compare tenths and then hundredths. 0.65 > 0.58.

7.26 Answer: **>**
Compare tenths and then hundredths. 0.75 > 0.68.

7.27 Answer: **>**
Compare tenths and then hundredths. 0.85 > 0.78.

7.28 Answer: **>**
Compare tenths and then hundredths. 0.95 > 0.88.

7.29 Answer: **>**
Compare tenths and then hundredths. 1.05 > 0.98.

7.30 Answer: **>**
Compare tenths and then hundredths. 1.15 > 1.08.

Chapter 8: Geometry

Concept: Points, lines, segments, rays, and parallel or perpendicular lines

What it means: A point shows an exact location. A line continues forever both ways, a segment has two endpoints, and a ray has one endpoint. Parallel lines never meet; perpendicular lines meet at right angles. **Remember:** Look for arrowheads and endpoints to name each figure.

Visual model

Questions

1. Which geometry word means a location?

2. Which geometry word means a location?

3. Which geometry word means two endpoints?

4. Which geometry word means two endpoints?

5. Which geometry word means one endpoint?

6. Which geometry word means one endpoint?

7. Which geometry word means lines that never meet?

8. Which geometry word means lines that never meet?

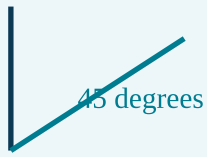
9. Which geometry word means lines meeting at a right angle?

10. Which geometry word means lines meeting at a right angle?

Concept: Measuring and classifying angles

What it means: An angle measures a turn. Acute angles are less than 90 degrees, right angles equal 90 degrees, and obtuse angles are between 90 and 180 degrees. **Example:** A 120-degree angle is obtuse. **Remember:** Compare the opening, not the length of the rays.

Visual model



acute < 90

Questions

11. Classify a 25-degree angle.

12. Classify a 45-degree angle.

13. Classify a 70-degree angle.

14. Classify a 90-degree angle.

15. Classify a 110-degree angle.

16. Classify a 135-degree angle.

17. Classify a 160-degree angle.

18. Classify a 85-degree angle.

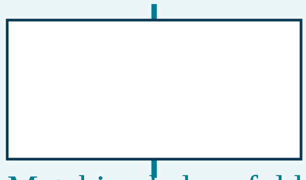
19. Classify a 100-degree angle.

20. Classify a 180-degree angle.

Concept: Triangles, quadrilaterals, and symmetry

What it means: Classify shapes by their sides and angles. A line of symmetry splits a figure into matching mirror-image halves. **Example:** A square has four equal sides, four right angles, and four lines of symmetry. **Remember:** Fold along a symmetry line; both sides should match exactly.

Visual model



line of symmetry

Matching halves fold together

Questions

21. How many lines of symmetry does a equilateral triangle have?

22. How many lines of symmetry does a square have?

23. How many lines of symmetry does a rectangle have?

24. How many lines of symmetry does a rhombus have?

25. How many lines of symmetry does a regular pentagon have?

26. How many lines of symmetry does a regular hexagon have?

27. How many lines of symmetry does a trapezoid have?

28. How many lines of symmetry does a scalene triangle have?

29. How many lines of symmetry does a kite have?

30. How many lines of symmetry does a parallelogram have?

Chapter Answers and Explained Solutions

Use these explanations to understand the concept and check how the answer is found.

8.1 Answer: **point**
A point is described as a location.

8.2 Answer: **point**
A point is described as a location.

8.3 Answer: **line segment**
A line segment is described as two endpoints.

8.4 Answer: **line segment**
A line segment is described as two endpoints.

8.5 Answer: **ray**
A ray is described as one endpoint.

8.6 Answer: **ray**
A ray is described as one endpoint.

8.7 Answer: **parallel lines**
A parallel lines is described as lines that never meet.

8.8 Answer: **parallel lines**
A parallel lines is described as lines that never meet.

8.9 Answer: **perpendicular lines**
A perpendicular lines is described as lines meeting at a right angle.

8.10 Answer: **perpendicular lines**
A perpendicular lines is described as lines meeting at a right angle.

8.11 Answer: **acute**
25 degrees is acute because it is less than 90 degrees.

8.12 Answer: **acute**
45 degrees is acute because it is less than 90 degrees.

8.13 Answer: **acute**
70 degrees is acute because it is less than 90 degrees.

8.14 Answer: **right**
90 degrees is right because it is equal to 90 degrees.

8.15 Answer: **obtuse**
110 degrees is obtuse because it is between 90 and 180 degrees.

8.16 Answer: **obtuse**
135 degrees is obtuse because it is between 90 and 180 degrees.

8.17 Answer: **obtuse**
160 degrees is obtuse because it is between 90 and 180 degrees.

8.18 Answer: **acute**
85 degrees is acute because it is less than 90 degrees.

8.19 Answer: **obtuse**
100 degrees is obtuse because it is between 90 and 180 degrees.

8.20 Answer: **straight**
180 degrees is straight because it is equal to 180 degrees.

8.21 Answer: **3**
A equilateral triangle has 3 line(s) of symmetry because only that many folds make matching mirror-image halves.

8.22 Answer: **4**
A square has 4 line(s) of symmetry because only that many folds make matching mirror-image halves.

8.23 Answer: **2**
A rectangle has 2 line(s) of symmetry because only that many folds make matching mirror-image halves.

8.24 Answer: **2**
A rhombus has 2 line(s) of symmetry because only that many folds make matching mirror-image halves.

8.25 Answer: **5**
A regular pentagon has 5 line(s) of symmetry because only that many folds make matching mirror-image halves.

8.26 Answer: **6**
A regular hexagon has 6 line(s) of symmetry because only that many folds make matching mirror-image halves.

8.27 Answer: **0**
A trapezoid has 0 line(s) of symmetry because only that many folds make matching mirror-image halves.

8.28 Answer: **0**
A scalene triangle has 0 line(s) of symmetry because only that many folds make matching mirror-image halves.

8.29 Answer: **1**
A kite has 1 line(s) of symmetry because only that many folds make matching mirror-image halves.

8.30 Answer: **0**
A parallelogram has 0 line(s) of symmetry because only that many folds make matching mirror-image halves.